

OS storage module, connections for network interfaces, and more.

3. **Mass Storage Device:** NAS heads could contain either HDDs or SSDs. There could be two to five storage drive bays. Finally, it uses RAID to arrange multiple physical storages as single or independent logical drives.
4. **Networking Interfaces:** NAS devices include a variety of interfaces for connecting to different networks. High-speed Wi-Fi, Ethernet LAN connections, USB ports, and other networking interfaces are the most popular ones found on NAS devices. Wi-Fi is used by a NAS head to configure the cloud. If not, you can attach the NAS to an intranet network via the Ethernet port. Additionally, you can connect laptops to the USB ports to set up the NAS, charge accessories, and other things.
5. **Operating Systems:** Modern, complex network-attached storage devices have a compact yet powerful operating system. The OS manages the CPU, RAM chips, network ports, and all other mass storage devices. These operating systems are preinstalled on the motherboard and include a NAS device out of the box. For these OSs, you typically don't need to perform any maintenance. The NAS maker configures cloud-based automated updates.

NAS Software Components

A NAS box must be controlled from a node computer linked to the network because it lacks a keyboard, mouse, and display.

To control the storage server on this node computer, you must visit the NAS programme. A NAS programme is a simple standalone programme or web application. Do note that although NASes use RAID technology, which offers fault-tolerance, they shouldn't be viewed as a backup alternative. This implies that you shouldn't keep a single duplicate of crucial data on a NAS.

How to Maintain NAS?

1. Understand the configuration

Write down the NAS configuration when you configure your NAS. In NAS configuration, there are:

- The number and size of discs
- The RAID type
- The filesystem type (if a NAS enables changing it)
- Any further deviations from the recommended setup

The number of discs may seem pointless to keep track of since it seems apparent, but if you ever update the NAS over time, the order of upgrades and the number of discs that were added or replaced with bigger ones become important.

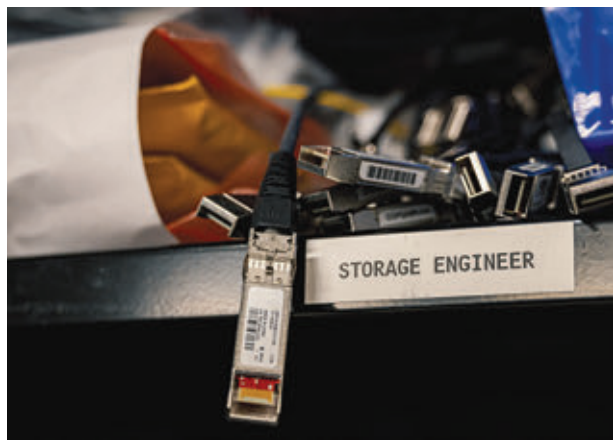
Additionally, it's a good idea to label your NAS's discs in accordance with their order. The top left disc is the first one since discs are numbered from left to right in a horizontal configuration and from top to bottom in a vertical layout.

Ask the person configuring your NAS to write down all the configuration settings on a piece of paper, then tape it to the NAS box.

All of this doesn't take long, but it would be quite helpful if NAS data recovery were necessary.

2. Monitor indicators and NAS performance

Always be on the lookout for any strange behaviour from your NAS. If the NAS, for instance, operates more slowly, it's possible that one of its member discs has failed and the system has transitioned to a degraded state. Data recovery is very difficult if a second disc fails before the first one is replaced. Pay attention to the disc condition indications as well. Check the NAS documentation to find out what this indicates and what you should do if the signs seem out of the ordinary.



3. Use a UPS (Uninterruptable Power Supply)

The rapid loss of power has a negative impact on NAS devices; each time you switch the NAS on after one of these occurrences, it starts a complete or partial rebuild. The rebuild considerably reduces NAS performance, and you risk losing some data due to a write hole.

Consider purchasing a UPS if you are in a region where power outages are frequent so that the NAS can properly shut down.

4. Disk Replacement

One of the key benefits of NAS is the ability to instantly replace a failing disc without having to shut down the system. Even though it seems so straightforward, the disc replacement step actually sees the majority of errors that result in data loss. Let's get into further depth about this.

One of the most frequent errors is failing to promptly replace a damaged disc. Despite the fact that many NASes alert users to disc issues via indicators and messages in NAS management software, many users frequently disregard all of the warnings and go on using the NAS. This is feasible because to the RAID technology, which enables access to the NAS's data even if one of the discs fails. Even when overall performance is declining,